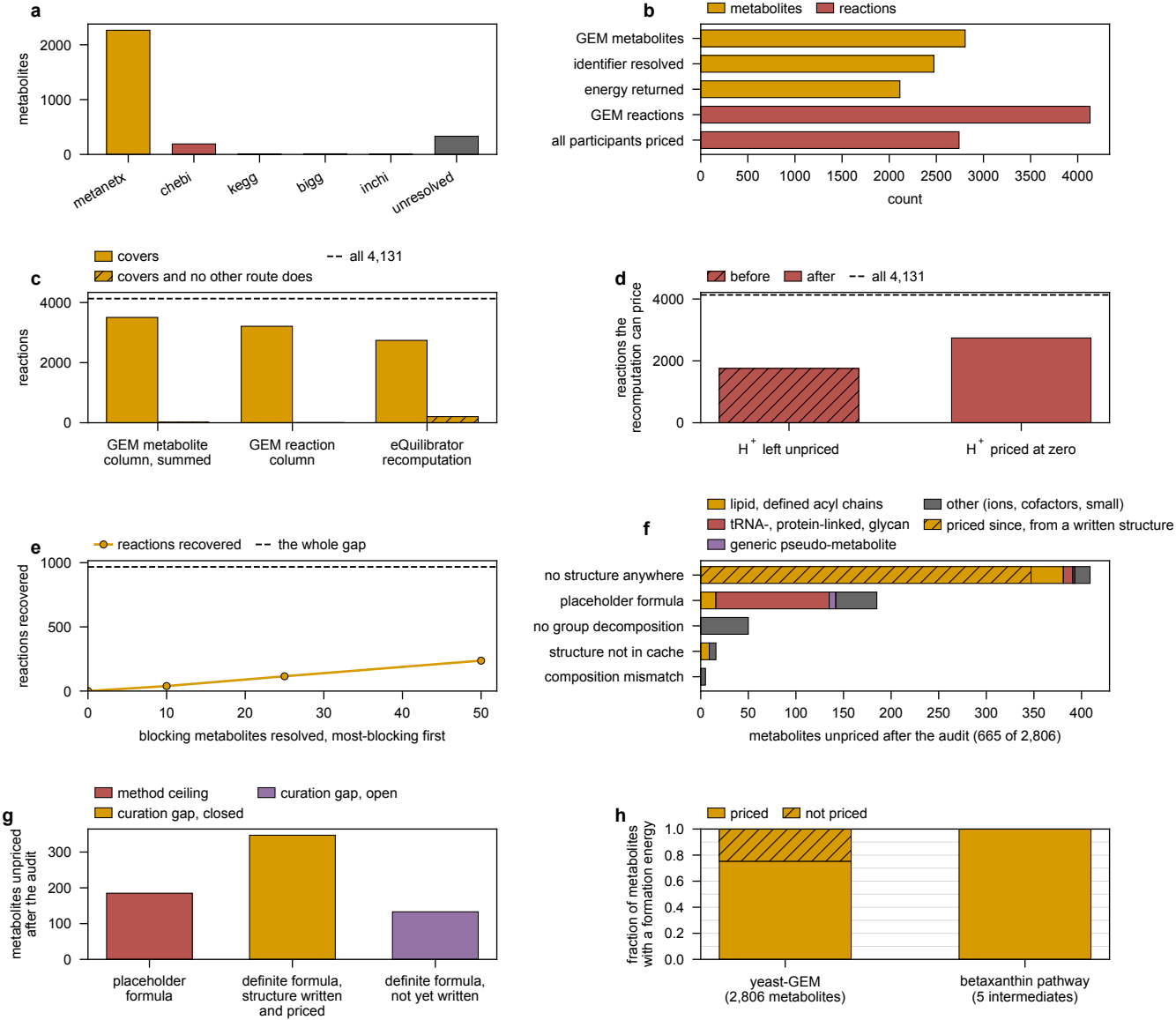
a
metabolites
2000
1000
0
metanetx
chebi
kegg
bigg
inchi
unresolved
b
metabolites
reactions
GEM metabolites
identifier resolved
energy returned
GEM reactions
all participants priced
0
500
1000
1500
2000
2500
3000
3500
4000
count
c
covers
covers and no other route does
all 4,131
reactions
4000
2000
0
GEM metabolite column, summed
GEM reaction column
eQuilibrator recomputation
d
before
after
all 4,131
reactions the recomputation can price
4000
2000
0
H+ left unpriced
H+ priced at zero
e
reactions recovered
the whole gap
reactions recovered
1000
500
0
0
10
20
30
40
50
blocking metabolites resolved, most-blocking first
f
lipid, defined acyl chains
other (ions, cofactors, small)
tRNA-, protein-linked, glycan
priced since, from a written structure
generic pseudo-metabolite
no structure anywhere
placeholder formula
no group decomposition
structure not in cache
composition mismatch
0
50
100
150
200
250
300
350
400
metabolites unpriced after the audit (665 of 2,806)
g
method ceiling
curation gap, open
curation gap, closed
metabolites unpriced after the audit
300
200
100
0
placeholder formula
definite formula, structure written and priced
definite formula, not yet written
h
priced
not priced
fraction of metabolites with a formation energy
1.0
0.8
0.6
0.4
0.2
0.0
yeast-GEM (2,806 metabolites)
betaxanthin pathway (5 intermediates)